Chpater4 Optics & Light

Lesson3 Double Plane, Slab, and Prism

1. A glass prism has an angle of prism $60°$. If the angle of minimum deviation is $40°$, find the refractive index of the glass.
2. A prism of small angle $5°$ is made of a material of refractive index 1.6. Calculate the angle of deviation.
3. A prism produces a minimum deviation of $30°$. The refractive index of the prism material is $\sqrt{3}$. Find the angle of prism.
4. The refractive index of glass prism is 1.5 and angle of prism is $10°$. Calculate minimum deviation.
5. Although we do not prove it here, the minimum angle of deviation δ_{min} for a prism occurs when the angle of incidence θ_1 is such that the refracted ray inside the prism makes the same angle with the normal to the two prism faces[1] as shown in Figure 35.17. Obtain an expression for the index of refraction of the prism material in terms of the minimum angle of deviation and the apex angle Φ.

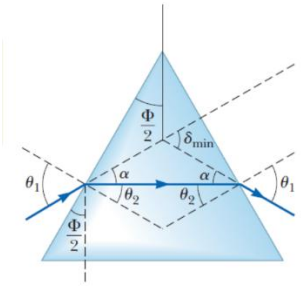


Figure 35.17 (Example 35.5) A light ray passing through a prism at the minimum angle of deviation δ_{min}.

6. **Figure P35.8** shows a refracted light beam in linseed oil making an angle of $\phi = 20.0°$ with the normal line NN'. The index of refraction of linseed oil is 1.48. Determine the angles (a) θ and (b) θ'.

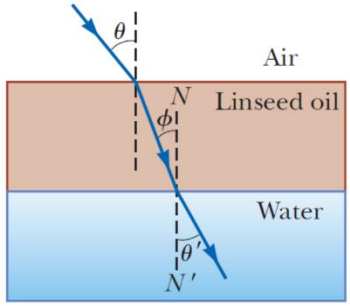


Figure P35.8

7. When the light ray illustrated in **Figure P35.22** passes through the glass block of index of refraction $n = 1.50$, it is shifted laterally by the distance d

   (a) Find the value of d

   (b) Find the time interval required for the light to pass through the glass block.

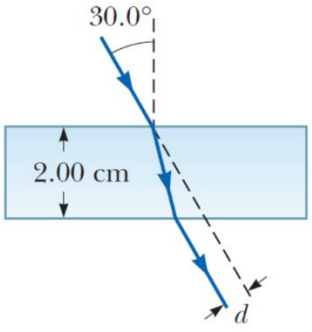


Figure P35.22

8. A small fish is swimming at a depth of 3.00 m below the surface of a pond. The refractive index of water is 1.33.

9. A triangular glass prism with apex angle $\Phi = 60.0^\circ$ has an index of refraction $n = 1.50$ (Fig. P35.43). What is the smallest angle of incidence θ_1 for which a light ray can emerge from the other side?

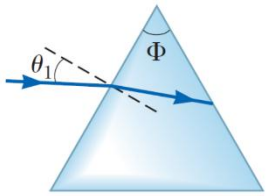


Figure P35.43

10. A beam of light is incident on a prism of a certain glass at an angle of $\theta_1 = 30.0^\circ$, as shown in **Figure 22.18.** If the index of refraction of the glass for violet light is 1.80, find **(a)** θ_2, the angle of refraction at the air–glass interface, **(b)** ϕ_2, the angle of incidence at the glass–air interface, and **(c)** ϕ_1, the angle of refraction when the violet light exits the prism. **(d)** What is the value of Δy, the amount by which the violet light is displaced vertically?

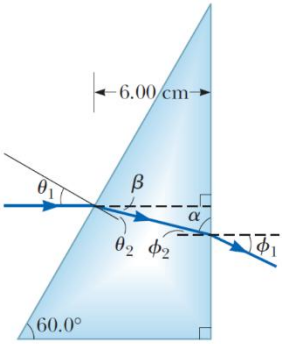


Figure 22.18
(Example 22.5)

11. The ray of light shown in **Figure 18-27** is incident upon a 60°-60°-60° glass prism, $n = 1.5$.

a. Using Snell's law of refraction, determine the angle, θ_2, to the nearest degree.

b. Using elementary geometry, determine the value of θ_1'.

c. Determine θ_2'.

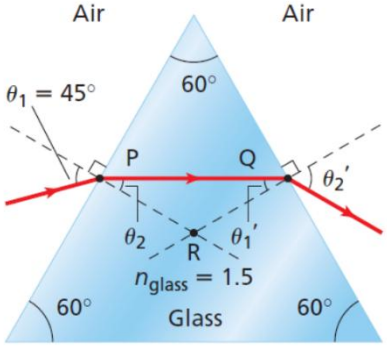


Figure 18-27

12. If the apex angle of a prism is $\phi = 75^\circ$ (see Fig. 23–63), what is the minimum incident angle for a ray if it is to emerge from the opposite side (i.e., not be totally internally reflected), given $n = 1.58$?

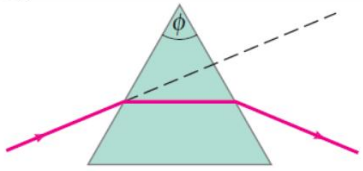


FIGURE 23–63
Problem 72.

13. We wish to determine the depth of a swimming pool filled with water by measuring the width ($x = 6.50$ m) and then noting that the far bottom edge of the pool is just visible at an angle of 13.0° above the horizontal as shown in **Fig. 23–62**. Calculate the depth of the pool.

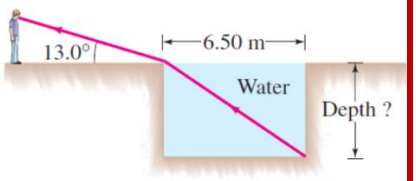


FIGURE 23–62
Problem 66.

14. Light is incident on an equilateral glass prism at a $45.0°$ angle to one face, **Fig. 23–64**. Calculate the angle at which light emerges from the opposite face. Assume that $n = 1.54$.

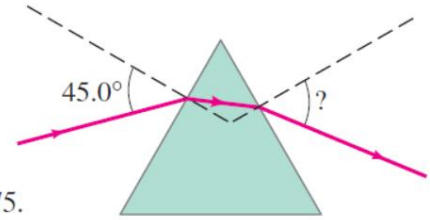


FIGURE 23–64
Problems 74 and 75.